

# VINAY KUMAR CHAKALI

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## EDUCATION

### Quinnipiac University — Master of Science, Computer Science

Hamden, CT • Expected May 2026

- **Relevant coursework:** Data Structures & Algorithms, Distributed Operating Systems, Database Systems, Cybersecurity, Software System Design (UML), Deep Learning, Cloud Computing & Microservices.
- **Certifications:** LangChain & RAG Systems (DeepLearning.AI) • Docker & Kubernetes for Developers (KodeKloud).

## WORK EXPERIENCE

### Starlite Infotech — Full-Stack Software Engineer

Bengaluru, India • Jan 2023 – Aug 2024

- Built full-stack features with **Python**, **React.js**, **TypeScript** and **Node.js**, delivering REST APIs and responsive UI components for core production modules.
- Improved key frontend screens for speed and accessibility, cutting initial page render time by **~25%** and reducing UI bugs caught in QA.
- Added **PyTest** and **Jest** coverage into the **CI/CD** pipeline, reducing bugs reaching production by **~35%**.
- Refactored **PostgreSQL** queries and added indexing to optimize database performance, lowering average API latency by **30%**.
- Containerized core services with **Docker**, improving deployment consistency and cutting page load times by **20%**.

## PROJECTS

### Polish — Music Matching Platform [\[GitHub\]](#)

Full-Stack Engineer (Personal Project) • Jun 2025

- Designed and built a real-time, high-concurrency music platform with a 30-second active-window matching engine using **FastAPI** and **WebSockets**.
- Tuned the matching algorithm to detect matches and deliver live messages in **under 4 seconds** for concurrent users.
- Secured the app with **JWT** auth and **bcrypt** password hashing, plus geolocation-based matching.
- Wrote 9 unit tests across auth and WebSocket flows to prevent regressions before deployment.

### Book Data Pipeline — Backend Service [\[GitHub\]](#)

Backend Engineer (Personal Project) • Jun 2025

- Built a data ingestion pipeline with **Python**, **Requests** and **BeautifulSoup** that automates book-catalog extraction, cutting manual gathering time by **98%**.
- Designed a normalized **SQLite** schema with **UNIQUE** constraints and **INSERT-OR-IGNORE** deduplication to keep data clean.
- Exposed the data through **FastAPI** REST endpoints with parameterized queries, server-side pagination, and fuzzy search — safe against SQL injection.

## TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL, HTML5, CSS3, C++
- **Frameworks:** React.js, Node.js, Express.js, FastAPI, SQLAlchemy, Pydantic, Tailwind CSS
- **AI/ML:** LLMs, RAG, LangChain, LangGraph, OpenAI API, Machine Learning
- **Tools & Infra:** Docker, Kubernetes, CI/CD, Git, GitHub, GitLab CI, Postman, Vite, PostgreSQL, SQLite, ChromaDB, Pinecone, WebSockets